

$S_t > \theta_t$ threshold crossing

Neural /
Thalamo-cortical
Substrate

S_t, θ_t
APGI
Variable(s)

EEG / MEG
Marker



Feedforward sensory input; error in early cortex

Neuromodulatory gating (interoceptive precision signals)

Recurrent loops in association cortex

Thalamic gate evaluates S_t vs. threshold θ_t

Cortico-thalamo-cortical broadcast via Global Workspace

Pred. error ϵ^l (local)
High $\rightarrow$ Low

Interoceptive Π^l (gain control)
Low $\rightarrow$ High

Saliency evidence S_t (accumulation)
Rises over time

Ignition if $S_t > \theta_t$ (threshold crossing)

Global evidence G_t (workspace stabilization)
Sustained High

Early Gamma / N1
~30-80 ms (sensory gain)

Alpha Suppression
~80-150 ms (attentional gain)

N2 / Sustained Gamma
~150-250 ms (accumulation phase)

P3b Onset
~250-320 ms (ignition)

Sustained Late Positivity
~320-500 ms (global broadcast)

0

50

120

250
Time (ms)

320

500

Phase (colour key)



1. Error Generation



2. Precision Weighting



3. S_t Accumulation



4. Ignition Decision



5. Global Broadcast